

NUMBER BASE SYSTEM

Introduction

A number base system is a method used to represent numbers using a set of digits or symbols.

The value of a number depends on:

- The digits used
- The position of the digits
- The base of the number system

The number system we use every day is called the decimal system or base ten system.

Computers, calculators, and digital devices also use other number systems such as binary, octal, and hexadecimal.

Meaning of Base or Radix

The base (or radix) of a number system is the number of different digits used in that system.

Examples:

- Base 10 uses ten digits: 0 – 9
- Base 2 uses two digits: 0 and 1
- Base 8 uses digits: 0 – 7
- Base 16 uses digits: 0 – 9 and A – F

Common Number Systems

Number System Base Digits Used

Binary	2	0, 1
Octal	8	0 – 7
Decimal	10	0 – 9
Hexadecimal	16	0 – 9 and A – F

In hexadecimal:

- A = 10
- B = 11
- C = 12
- D = 13
- E = 14
- F = 15

Place Value in Number Bases

The value of a digit depends on its position.

Example in base 10:

$$352_{10}$$

$$= (3 \times 10^2) + (5 \times 10^1) + (2 \times 10^0)$$

$$= (3 \times 100) + (5 \times 10) + (2 \times 1)$$

$$= 300 + 50 + 2$$

$$= 352$$

Expanded Form of Numbers in Bases

Any number can be expanded using powers of its base.

Example 1

Convert 231_4 to base ten.

Step 1:

Expand the number.

$$231_4$$

$$= (2 \times 4^2) + (3 \times 4^1) + (1 \times 4^0)$$

Step 2:

Calculate.

$$= (2 \times 16) + (3 \times 4) + (1 \times 1)$$

$$= 32 + 12 + 1$$

$$= 45$$

Therefore:

$$231_4 = 45_{10}$$

Converting Base Ten Numbers to Other Bases

This is done using repeated division.

Example 2

Convert 25_{10} to base 2.

Step 1:

Divide repeatedly by 2.

$$25 \div 2 = 12 \text{ remainder } 1$$

$$12 \div 2 = 6 \text{ remainder } 0$$

$$6 \div 2 = 3 \text{ remainder } 0$$

$$3 \div 2 = 1 \text{ remainder } 1$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Step 2:

Read remainders from bottom to top.

$$11001_2$$

Therefore:

$$25_{10} = 11001_2$$

Example 3

Convert 57_{10} to base 8.

$$57 \div 8 = 7 \text{ remainder } 1$$

$$7 \div 8 = 0 \text{ remainder } 7$$

Read upward:

$$71_8$$

Therefore:

$$57_{10} = 71_8$$

Converting Other Bases to Base Ten

Multiply each digit by the appropriate power of the base.

Example 4

Convert 1011_2 to base ten.

$$1011_2$$

$$= (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

$$= 8 + 0 + 2 + 1$$

$$= 11$$

Therefore:

$$1011_2 = 11_{10}$$

Example 5

Convert 243_5 to base ten.

$$243_5$$

$$= (2 \times 5^2) + (4 \times 5^1) + (3 \times 5^0)$$

$$= (2 \times 25) + (4 \times 5) + (3 \times 1)$$

$$= 50 + 20 + 3$$

$$= 73$$

Therefore:

$$243_5 = 73_{10}$$

Conversion from One Base to Another

Convert first to base ten, then to the required base.

Example 6

Convert 213_4 to base 2.

Step 1:

Convert to base ten.

$$213_4$$

$$= (2 \times 4^2) + (1 \times 4^1) + (3 \times 4^0)$$

$$= 32 + 4 + 3$$

$$= 39_{10}$$

Step 2:

Convert 39_{10} to base 2.

$$39 \div 2 = 19 \text{ remainder } 1$$

$$19 \div 2 = 9 \text{ remainder } 1$$

$$9 \div 2 = 4 \text{ remainder } 1$$

$$4 \div 2 = 2 \text{ remainder } 0$$

$$2 \div 2 = 1 \text{ remainder } 0$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Reading upward:

$$100111_2$$

Therefore:

$$213_4 = 100111_2$$

Addition in Number Bases

Addition is similar to ordinary addition but carried out according to the base.

Example 7

Add 101_2 and 111_2 .

$$\begin{array}{r} 101 \\ + 111 \\ \hline \end{array}$$

$$1100$$

Therefore:

$$101_2 + 111_2 = 1100_2$$

Subtraction in Number Bases

Borrowing is done according to the base.

Example 8

Subtract 1011_2 from 1110_2 .

1110

• 1011

11

Therefore:

$$1110_2 - 1011_2 = 11_2$$

Multiplication in Number Bases

Example 9

Multiply 11_2 by 10_2 .

11

× 10

00

+ 110

110

Therefore:

$$11_2 \times 10_2 = 110_2$$

Binary Numbers

Binary numbers are numbers expressed in base two.

Digits used:

- 0
- 1

Binary numbers are very important in computers and digital electronics.

Examples:

- 101_2
 - 1101_2
 - 111111_2
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Octal Numbers

Octal numbers use base eight.

Digits:

0 – 7

Example:

- 127_8
 - 345_8
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Hexadecimal Numbers

Hexadecimal uses base sixteen.

Digits:

0 – 9 and A – F

Examples:

- $2A_{16}$
 - $7F_{16}$
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Uses of Number Bases

1. Computers use binary numbers.
 2. Digital electronics operate in binary.
 3. Hexadecimal is used in programming.
 4. Octal simplifies long binary numbers.
 5. Number bases are useful in coding and data representation.
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Important Notes

1. Digits in a base must always be less than the base.

Correct:

- 101_2
- 237_8

Incorrect:

- 245_2
 - 89_8
2. The rightmost digit has power zero.
 3. Conversion between bases is a common examination question.
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Worked Examples

Example 10

Convert 110101_2 to base ten.

$$= (1 \times 2^5) + (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$

$$= 32 + 16 + 0 + 4 + 0 + 1$$

$$= 53$$

Therefore:

$$110101_2 = 53_{10}$$

Example 11

Convert 96_{10} to base 2.

$$96 \div 2 = 48 \text{ remainder } 0$$

$$48 \div 2 = 24 \text{ remainder } 0$$

$$24 \div 2 = 12 \text{ remainder } 0$$

$$12 \div 2 = 6 \text{ remainder } 0$$

$$6 \div 2 = 3 \text{ remainder } 0$$

$$3 \div 2 = 1 \text{ remainder } 1$$

$$1 \div 2 = 0 \text{ remainder } 1$$

Reading upward:

1100000_2

Therefore:

$$96_{10} = 1100000_2$$

Practice Questions

1. Convert 101101_2 to base ten.
 2. Convert 145_{10} to base 2.
 3. Convert 243_5 to base ten.
 4. Convert 75_{10} to base 8.
 5. Convert $2A_{16}$ to base ten.
 6. Add:
 $1011_2 + 111_2$
 7. Subtract:
 $1101_2 - 101_2$
 8. Convert 321_4 to base ten.
 9. Convert 111111_2 to base ten.
 10. Convert 64_{10} to base 2.
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Conclusion

A number base system is a way of representing numbers using different sets of digits. Understanding number bases helps students perform conversions, solve arithmetic operations in different bases, and understand how computers process information. Mastery of binary, octal, decimal, and hexadecimal systems is important in mathematics, computer science, and technology.